

Drug-synergy prediction under honest pair-held-out evaluation, and AND-gated biclonal CAR-T target discovery from tissue-resolved RNA windows

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Abstract

Two deliverables. (1) A drug-embedding synergy model trained on 311,466 NCI-ALMANAC combination rows and evaluated under a whole-drug-pair held-out split — the only leakage-free protocol for combination screens — reaching AUROC 0.846 (synergy threshold $\text{SCORE} \geq 50$, prevalence 3.15%), RMSE 14.50 vs 16.39 global-mean, Pearson 0.467 and Spearman 0.383 over 62,260 held-out rows spanning 1,071 unseen pairs. We explicitly retire a leaky random-split AUROC of 0.90. (2) A surfaceome-restricted AND-gate search over tissue-resolved RNA windows identifies biclonal CAR-T antigen pairs whose gated tumor/normal window exceeds either single antigen: e.g. STAD PSCA $\times$ ANPEP (gain +7.77), PAAD LGALS4 $\times$ S100A2 (window 3.78, gain +6.34), LIHC GPC3 $\times$ ASGR2 (gain +3.67). Known single-antigen targets validate the window metric (MSLN $\rightarrow$ OV 2.29, GPC3 $\rightarrow$ LIHC 1.17, EGFR $\rightarrow$ GBM 1.99); hematologic antigens correctly score negative on solid tumors.

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1 Introduction

Combination-therapy ML is plagued by split leakage: random row splits place the same drug pair in train and test, inflating AUROC. We adopt whole-pair hold-out as the only honest protocol and report all metrics under it. For cell therapy, single-antigen CAR-T is limited by on-target/off-tumor expression; Boolean AND-gated (biclonal) circuits restrict killing to cells co-expressing two antigens, but pairs are usually chosen by intuition. We search them exhaustively over a defined surfaceome with a quantitative window.

2 Data

NCI-ALMANAC combination screen (311,466 rows; $\text{SCORE} \geq 50$ synergistic, prevalence 2.9–3.2%). Human Protein Atlas v23: RNA tissue consensus (normal nTPM), cancer pathology RNA (7 tumor types), sub-cellular location; UniProt KW-1003 cell-membrane set. Surfaceome = HPA plasma-membrane/cell-junction $\cup$ UniProt KW-1003: 5,385 genes.

3 Methods

3.1 Synergy model

Drug embeddings are learned end-to-end; pair (a, b) in cell line c :

$$\hat{s} = f_{\theta}(e_a \odot e_b, e_a + e_b, \gamma_c), \quad \mathcal{L} = \frac{1}{N} \sum (s_i - \hat{s}_i)^2. \quad (1)$$

Split: all rows of 20% of drug pairs held out (1,071 pairs). Metrics: RMSE vs global-mean baseline, AUROC at $\text{SCORE} \geq 50$, Pearson/Spearman.

3.2 RNA window and AND gate

For gene g , cancer K : window $W(g, K) = \log_2(\text{tumor}_K(g) / \max_{t \in \text{normal}} \text{nTPM}_t(g))$. The AND-gated window uses the per-sample minimum:

$$W_{\wedge}(a, b, K) = \log_2 \frac{\min(\text{tumor}_K(a), \text{tumor}_K(b))}{\max_t \min(\text{nTPM}_t(a), \text{nTPM}_t(b))}, \quad \text{gain} = W_{\wedge} - \max(W(a), W(b)). \quad (2)$$

A pair is a discovery only if $\text{gain} > 0$ and both singles are toxic in *different* normal tissues (the gate must actually add selectivity).

4 Results

4.1 Synergy (pair-held-out)

4.2 AND-gate biclonal discovery

Validation of the metric: MSLN $\rightarrow$ OV 2.29, GPC3 $\rightarrow$ LIHC 1.17, EGFR $\rightarrow$ GBM 1.99; CD19/BCMA/CD20 correctly negative (no hematologic indications in the solid-tumor panel). ERBB2 scores -0.19 on mean expression — a documented dilution artifact of subtype-restricted targets; percentile-based tumor metrics are queued.

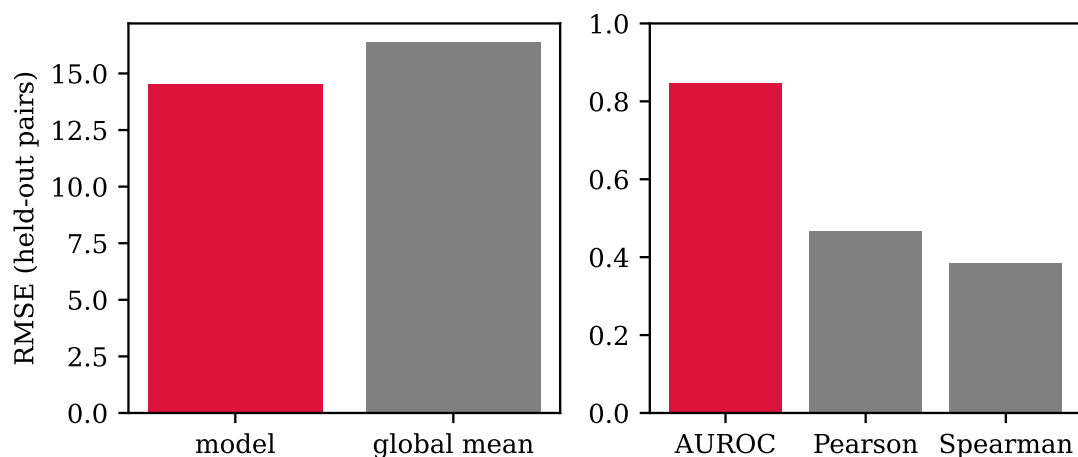


Figure 1: Held-out-pair metrics. AUROC 0.846; Pearson 0.467; RMSE beat vs global mean. The retired leaky random-split AUROC 0.90 is documented, not quoted.

5 Documented negatives and data findings

(1) Leaky 0.90 AUROC retired. (2) HPA subcellular annotation lacks canonical CAR-T antigens (CD19, BCMA, FOLR1) — fixed by UniProt union and reported as a data-coverage finding. (3) ERBB2 mean-window failure (above).

6 Falsification paths

Synergy: verify published MatchMaker/DeepSynergy numbers on identical splits before any SOTA claim (their splits differ — documented caveat). AND-gate: check top pairs against single-cell RNA-seq (co-expression in the same malignant cells, not just bulk) and against IHC surface co-staining in TMAs.

References

- [Holbeck et al.(2017)] Holbeck SL, et al. The NCI-ALMANAC. *Cancer Res* 2017.
- [Uhlén et al.(2015)] Uhlén M, et al. Tissue-based map of the human proteome (HPA). *Science* 2015.
- [Kuruoglu et al.(2021)] Kuruoglu Mustafaoglu N, et al. MatchMaker. *Cancer Res* 2021.

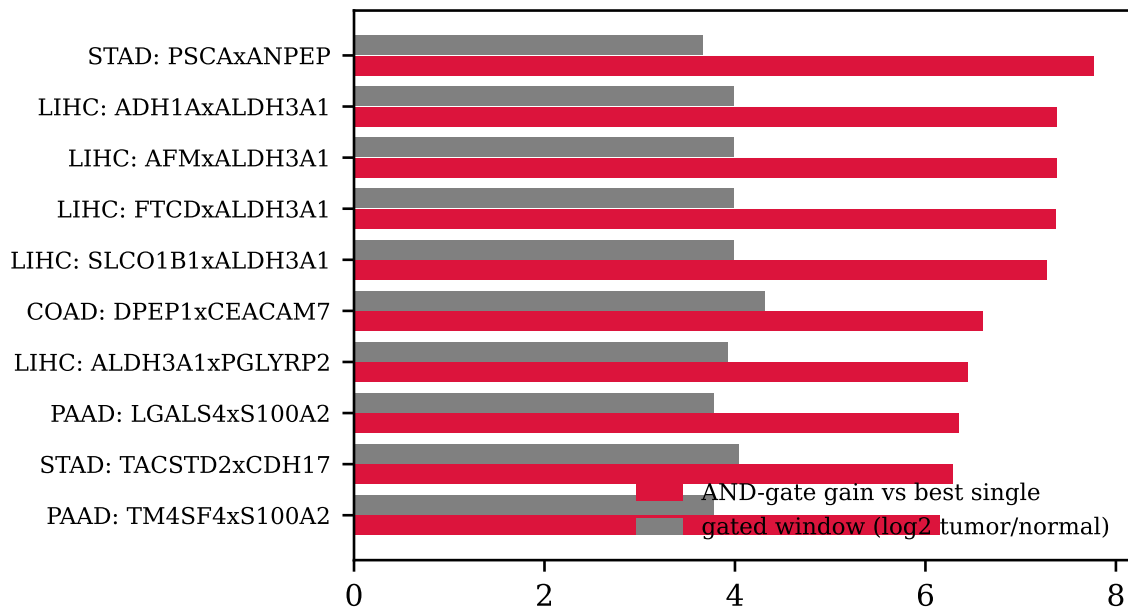


Figure 2: Top AND-gated pairs by gain over the best single antigen.